2nd call of Combinatorics & Probability **Univ. Rovira i Virgili**
February 1, 2024 **Version A**

You have 2 hours for completing the exam.
No electronic devices are allowed (i.e. calculators, mobile phones, etc.).
All answers must include the supporting deductions and calculations (if the calculations are too heavy then they can be left implied in the final answer).
If some answer does not include all the necessary information, it will be deemed invalid.

Name and surname:

1. (Total: 2 points) Prove that $n^3 - n$ is divisible by 6 for every $n \in \mathbb{N}$.

   Hint: Use mathematical induction, and analyze the different cases that may arise.

   Solution

   Induction base:

   Case $n = 1$: $1^3 - 1 = 0$, which is divisible by 6.

   Case $n = 2$: $2^3 - 2 = 6$, which is also divisible by 6.

   Induction hypothesis:

   We assume that the property holds for every positive integer $k \leq n$.

   Induction step:

   Now we prove that the property holds for $n+1$. In other words, we have to prove that $(n+1)^3 - (n+1)$ is a multiple of 6. Let us calculate the difference between $(n+1)^3 - (n+1)$ and $(n-1)^3 - (n-1)$:

   $$\Big[(n+1)^3 - (n+1)\Big] - \Big[(n-1)^3 - (n-1)\Big] = 6n^2$$

   In other words,

   $$(n+1)^3 - (n+1) = (n-1)^3 - (n-1) + 6n^2,$$

   By the induction hypothesis we know that $(n-1)^3 - (n-1)$ is a multiple of 6, and $6n^2$ is clearly also a multiple of 6, hence $(n+1)^3 - (n+1)$ is a multiple of 6, as desired.

There is another nice proof that does not use induction. We can easily verify that $n^3 - n = n(n-1)(n+1)$. In other words, $n^3 - n$ is the product of three consecutive integers. Among these three consecutive integers, at least one of them must be even (i.e. a multiple of 2), and exactly one of them must be a multiple of 3. Hence, $n^3 - n$ is a multiple of 2 and 3 at the same time, and hence it is a multiple of 6.

Q.E.D.

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2. (Total: 2 points) Suppose we want to distribute 11 distinct books among 4 students.

   (a) (1 point) Find the number of possible distributions so that every student gets at least two books.

   (b) (1 point) Find the number of possible distributions so that every student gets at least one book, and all the students get a different number of books.

Solution

(a) The number of was to distribute 11 distinguishable books into 4 distinguishable boxes (students), so that the first student gets 5 books, and the other three get 2 books is

$$\binom{11}{5,2,2,2} = 41\ 580$$

(See Examples 40 and 41, page 26, *Apuntes de Matemática Discreta*, de Juan A. Rodríguez Velázquez)

Now, we have 4 different ways to arrange the numbers $(5,2,2,2)$, all yielding the same result, by the symmetry of multinomial coefficients:

$$\binom{11}{5,2,2,2} = \binom{11}{2,5,2,2} = \binom{11}{2,2,5,2} = \binom{11}{2,2,2,5}.$$

Another possibility is to distribute the books as follows:

$$\binom{11}{4,3,2,2} = 69\ 300$$

There are $\dfrac{4!}{2!} = \dfrac{24}{2} = 12$ distinct permutations of the four numbers $(4,3,2,2)$.

Similarly, we have 4 different ways to arrange the numbers $(3,3,3,2)$, and

$$\binom{11}{3,3,3,2} = 92\ 400$$

Therefore, there is a total of

$$\begin{aligned} 4\binom{11}{5,2,2,2} + 12\binom{11}{4,3,2,2} + 4\binom{11}{3,3,3,2} &= 41\ 580 + 69\ 300 + 92\ 400 \\ &= 203\ 280 \end{aligned}$$

ways to distribute the books.

Note: There is no need to provide the numerical result.

(b) Following the same method as in the previous question we can see that there is only one combination of different numbers that add up to 11, namely $(1,2,3,5)$. Therefore we have

$$24\binom{11}{1,2,3,5} = 24 \cdot 27\ 720 = 665\ 280$$

ways to distribute the books.

Note: There is no need to provide the numerical result.

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3. (Total: 2 points) A bag contains 100 coins, where all of them are fair except one, which has HEADS on both sides.

   (a) (1 point) We extract a coin at random from the bag and toss it (we are not allowed to verify the coin before tossing it). What is the probability of getting HEADS?

   (b) (1 point) Now we extract a coin at random from the bag and toss it n times, getting HEADS all n times. Again, we are not allowed to verify the coin before tossing it, we can only see the result of each toss. Calculate the probability that the extracted coin is the unfair coin.

Solution

(a) We define the events

- $B \equiv$ "The extracted coin is the biased coin",
- $H \equiv$ "We get HEADS in one toss".

We have two cases: Either the extracted coin is fair, or it is the biased coin (i.e. $\overline{B}$ or B). By the formula of total probability we get

$$\begin{aligned}\mathbb{P}(H) &= \mathbb{P}(H|B)\ \mathbb{P}(B) + \mathbb{P}(H|\overline{B})\ \mathbb{P}(\overline{B}) \\ &= 1 \cdot \frac{1}{100} + \frac{1}{2} \cdot \frac{99}{100} = \frac{2+99}{200} = \\ &= \frac{101}{200} = 0,505.\end{aligned}$$

(b) We now define another event, namely $N \equiv$ "We get n HEADS in n tosses". First we will calculate $\mathbb{P}(N)$ with the formula of total probability, as in the previous example.

If we pick the biased coin and toss it n times we will get HEADS all n times. On the other hand, if we pick a fair coin and toss it n times, the probability of getting HEADS n times is $\left(\frac{1}{2}\right)^n$. Thus,

$$\begin{aligned}\mathbb{P}(N) &= \mathbb{P}(N|B)\ \mathbb{P}(B) + \mathbb{P}(N|\overline{B})\ \mathbb{P}(\overline{B}) \\ &= 1 \cdot \frac{1}{100} + \left(\frac{1}{2}\right)^n \cdot \frac{99}{100} = \frac{2^n+99}{100 \cdot 2^n}\end{aligned}$$

Now, using Bayes' rule we get

$$\begin{aligned}\mathbb{P}(B|N) &= \frac{\mathbb{P}(N|B)\ \mathbb{P}(B)}{\mathbb{P}(N)} = \frac{\mathbb{P}(N|B)\ \mathbb{P}(B)}{\mathbb{P}(N|B)\ \mathbb{P}(B) + \mathbb{P}(N|\overline{B})\ \mathbb{P}(\overline{B})} \\ &= \frac{1 \cdot \frac{1}{100}}{\frac{2^n+99}{100 \cdot 2^n}} = \frac{2^n}{2^n+99}\end{aligned}$$

Note that $\mathbb{P}(B|N)$ approaches 1 as n grows.

4. (Total: 2 points) The company FLIRTBUS operates buses with a capacity of 20 seats, but it usually sells 25 tickets for every bus, knowing that the probability that a random passenger shows up for the trip is $0,9$. Let χ be a random variable that indicates the number of passengers of a given bus who actually show up for the trip.

   (a) (1 point) Calculate the probability $\mathbb{P}(\chi \leq 20)$ (i.e. the probability that there will be no passengers left).

(b) (1 point) The company earns 20 Euros for every ticket sold, after the corresponding deductions. However, if more than 20 passengers show up for a given trip, the company has to charter an extra bus to accommodate the extra passengers, which has a cost of $4\ 000$ Euros. Is the company losing money in the long term? What is the average balance?.

Solution

(a) The random variable χ has a binomial distribution with parameters $n = 25$, $p = 0,9$ and $q = 0,1$. Hence

$$\begin{aligned}
\mathbb{P}(\chi \leq 20) &= \sum_{k=0}^{20} (\chi = k) = \sum_{k=0}^{20} \binom{n}{k} p^k q^{n-k} \\
&= 1 - \mathbb{P}(\chi \geq 21) = 1 - \sum_{k=21}^{25} \binom{n}{k} p^k q^{n-k} \\
&= 1 - \left[\binom{25}{21} p^{21} q^4 + \binom{25}{22} p^{22} q^3 + \right. \\
&\quad \left. + \binom{25}{23} p^{23} q^2 + \binom{25}{24} p^{24} q^1 + \binom{25}{25} p^{25} q^0 \right] = \\
&= 1 - \left[\binom{25}{4} p^{21} q^4 + \binom{25}{3} p^{22} q^3 + \right. \\
&\quad \left. + \binom{25}{2} p^{23} q^2 + \binom{25}{1} p^{24} q^1 + \binom{25}{0} p^{25} q^0 \right] = \\
&= 1 - \Big[12650\ (0,9)^{21} (0,1)^4 + 2300\ (0,9)^{22} (0,1)^3 + \\
&\quad + 300\ (0,9)^{23} (0,1)^2 + 25\ (0,9)^{24} (0,1) + (0,9)^{25} \Big] \\
&= 0,902006.
\end{aligned}$$

In the exam it is not necessary to give the exact numeric answer.

(b) The probability that the company has to charter an extra bus is

$$\begin{aligned}
\mathbb{P}(\chi > 20) = \mathbb{P}(\chi \geq 21) &= \sum_{k=21}^{25} \binom{n}{k} p^k q^{n-k} = \\
&= \left[\binom{25}{4} p^{21} q^4 + \binom{25}{3} p^{22} q^3 + \binom{25}{2} p^{23} q^2 + \right. \\
&\quad \left. + \binom{25}{1} p^{24} q^1 + \binom{25}{0} p^{25} q^0 \right] = 0,097994.
\end{aligned}$$

Now, in order to calculate the expected (or average) balance B of the company in Euros we do the following

$$\begin{aligned} B &= 25 \cdot 20 - \mathbb{P}(\chi > 20) \cdot 4000 \\ &= 500 - 391,976 = 108,024. \end{aligned}$$

As we can see, the balance is positive. Therefore, the company is earning money in the long term.

Note: Since calculators are not allowed in the exam, the student is only expected to write down the above formulas, without giving a numerical result. And of course, it is very difficult to answer the question whether the company is losing money without having a numerical result.

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5. (Total: 2 points) Let χ be a continuous random variable with Probability Density Function (PDF)

$$f(x) = \frac{c}{1+x^2}$$

for all $x \in \mathbb{R}$.

(a) (1 point) Determine the value of c and the Cumulative Distribution Function (CDF) $F(x)$ of χ.

(b) (1 point) Compute the variance of χ (if it exists).

Solution

(a) We have that

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi,$$

therefore $c = \dfrac{1}{\pi}$, and

$$F(x) = \frac{1}{\pi} \int_{-\infty}^{x} \frac{dt}{1+t^2} = \frac{1}{\pi} \arctan x + \frac{1}{2}$$

(b) In order to calculate $\mathbb{V}(\chi)$ first we must calculate $\mathbb{E}(\chi)$:

$$\mathbb{E}(\chi) = \int_{-\infty}^{\infty} t\ f_\chi(t)\ dt = \frac{1}{\pi} \int_{\infty}^{\infty} \frac{t\ dt}{1+t^2}.$$

However, the above integral does not converge, therefore, there can be no expected value, and hence, no variance.

Nevertheless, looking at the symmetry of the PDF, one might be tempted to conclude that $\mathbb{E}(\chi) = 0$. Even if we do that, we get to the integral

$$\mathbb{V}(\chi) = \int_{-\infty}^{\infty} (t-0)^2 \, f_\chi(t) \, dt = \frac{1}{\pi} \int_{\infty}^{\infty} \frac{t^2 \, dt}{1+t^2},$$

which does not converge either.